

ACCREDIX — SYSTEM ARCHITECTURE & TECHNICAL GUIDE

AI-Powered Continuous Accreditation Readiness System for NBA Criteria 3, 4, 5 & 6

1. Executive Summary & Core Concept

ACCREDIX is an intelligent, continuous accreditation-readiness platform designed to solve the core bottleneck of National Board of Accreditation (NBA) compliance in higher education institutions. Unlike traditional, static college ERPs that function merely as data storage portals, ACCREDIX operates as a **Digital Twin Accreditation Readiness Engine**. It continuously collects department metrics, cross-verifies evidence documents, performs real-time gap analysis, suggests actionable remedial steps, dynamically recalculates attainment scores, and executes Virtual NBA Committee Inspection simulations.

2. Continuous Accreditation Lifecycle (6-Step Engine)

Stage	Process Name	Technical Function & Module
1	COLLECT	Ingests course outcome evaluations, student grades, faculty portfolios, and publication files.
2	STORE	Maintains reactive, centralized application state (`appState`) and structured document index.
3	CHECK	Evaluates actual metrics against NBA compliance rules (e.g. CO target 70%, SFR threshold 1:15–1:25).
4	ANALYZE	Identifies weak outcomes, low attendance risks, unverified faculty proofs, and missing documents.
5	FIX	Generates targeted Action Items (`ACT-101`) with assigned owners, deadlines, and required evidence.
6	PROVE	Constructs verified Evidence Chains and executes Virtual NBA Inspection simulations.

3. Technology Stack & Architectural Constraints

- **Pure Native Stack:** Built 100% with standard HTML5, CSS3, and ES6+ Vanilla JavaScript.
- **Zero Framework Overhead:** No React, Vue, Angular, or TypeScript runtime overhead. Enables instant execution.
- **Zero Build Tools Needed:** Runs directly on any web server or static host (`python -m http.server`).
- **Single Page Application (SPA) Router:** Seamless tab routing without page reloads using DOM view toggles.
- **Reactive UI State Engine:** Dynamic DOM re-rendering triggered on state mutations in `appState`.

4. NBA Criteria Functional Deep-Dive

Criterion 3 — Course Outcomes & Program Outcomes (CO/PO):

Measures learning outcome attainment across courses. Includes Course Outcome attainment progress bars, direct & indirect assessment question mapping with Bloom's Taxonomy classification (Understand, Apply, Analyze, Evaluate), and a color-coded 8x4 CO-PO mapping matrix heatmap.

Criterion 4 — Student Performance & Early Intervention:

Monitors student academic progression, attendance rates, internal test scores, backlogs, and co-curricular participation. Computes an automated early intervention risk index categorizing students into *ON_TRACK*, *NEEDS_MONITORING*, or *HIGH_ATTENTION* profiles.

Criterion 5 — Faculty Information & Cadre Ratios:

Audits faculty qualifications (PhD vs Master's), teaching experience, weekly teaching workload, profile document completeness, and Student-Faculty Ratio (SFR). Highlight missing compliance records such as appointment orders or PhD registration certificates.

Criterion 6 — Faculty Contributions & Research Productivity:

Tracks research publications in Scopus/SCI indexed journals, patent grants, funded projects, and FDP certifications. Features an **Evidence Chain Inspector** linking claims to verified proof files.

5. Complete File Structure & Directory Guide

File / Path	Role / Purpose	Key Responsibilities
`frontend/index.html`	Application Shell	Main SPA HTML layout, header, search bar, modals, and script tags.
`frontend/src/styles.css`	Design Tokens & UI Styles	CSS variables, dark theme, glassmorphism panels, CO bars, and responsiveness.
`frontend/src/app.js`	State & View Controller	Central `appState`, tab switching, dynamic score calculation, and action resolution.
`frontend/src/utils/utils.js`	Shared Utilities	`Utils.showToast()` notifications, data formatters, and DOM helpers.
`src/components/Sidebar/sidebar.js`	Sidebar Component	Renders active navigation sidebar links and badge counts.
`src/components/Charts/charts.js`	Chart Component	SVG trend lines and distribution charts.
`src/components/DashboardCard/*`	KPI Card Component	Renders standard KPI metric cards with trends.
`src/components/DataTable/*`	Data Table Component	Standardized table renderers with search & filter hooks.
`src/components/FileUpload/*`	Upload Component	AI document extraction simulation & preview panel.
`src/components/StatusBadge/*`	Status Chip Component	Renders status indicators (PASS, WARNING, CRITICAL).
`src/pages/Dashboard/*`	Dashboard Module	Department readiness overview & live audit log timeline.
`src/pages/Criterion3/*`	Criterion 3 Module	CO/PO assessment views, Bloom's mapping & heatmaps.
`src/pages/Criterion4/*`	Criterion 4 Module	Student performance directory & early risk alerts.
`src/pages/Criterion5/*`	Criterion 5 Module	Faculty cadre audit, SFR calculator & workload charts.

`src/pages/Criterion6/*`	Criterion 6 Module	Publication ledger & Evidence Verification Chain.
`src/pages/GapAnalysis/*`	Gap Analysis Module	Action item management & dynamic gap resolution engine.
`src/pages/Evidence/*`	Evidence Vault Module	Digital twin document repository & verified proof index.
`src/pages/VirtualInspection/*`	Inspection Simulator Module	Virtual NBA committee inspection engine & SAR score audit.
`src/pages/Reports/*`	SAR Reports Module	PDF report exporter & executive summary generator.
`src/pages/AIAssistant/*`	AI Assistant Module	NBA SAR compliance Q&A; chat assistant.
`src/pages/Login/*`	Login Module	Access control interface for HOD & Faculty roles.

6. How to Run the System

Open PowerShell or Terminal in the project root directory and run:

```
cd frontend
```

```
python -m http.server 5173 --bind 127.0.0.1
```

Access the web platform in any modern browser at: **<http://127.0.0.1:5173>**